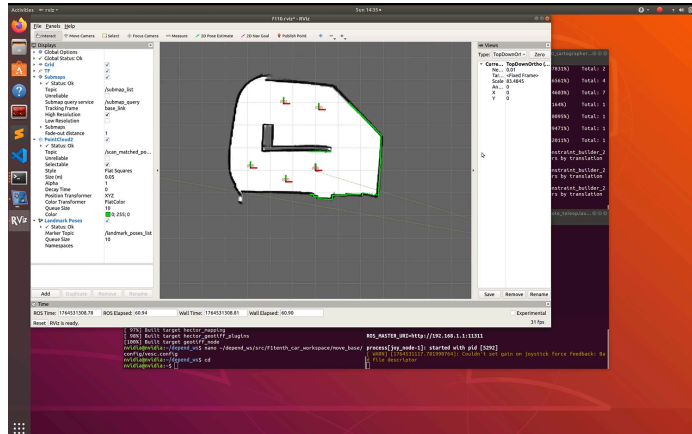
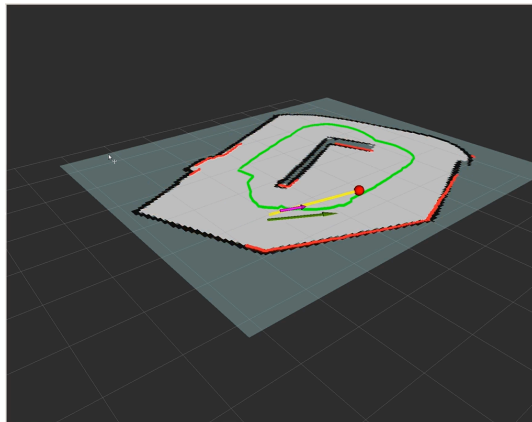


Group 2 PDF Report

Q1: ROS Cartographer SLAM Map



Q2: Car Localized by Particle Filter With Reference Path



Q3: Lookahead Distance Explanation

The value is calculated using $L = L_{\min} + ((v - V_{\min}) / (V_{\max} - V_{\min})) * (L_{\max} - L_{\min})$, which gets the ratio between the minimum and maximum lookahead based on current speed.

We selected $L_{\min} = 0.8$ (lookahead at low speed) and $L_{\max} = 2.0$ (lookahead at high speed). Similarly, $V_{\min} = 8.0$ defines the speed where the minimum lookahead should be used, and $V_{\max} = 15.0$ defines the speed where the maximum lookahead should be used. These values were tuned through testing to achieve stable straight-line behavior while maintaining good corner tracking without oscillations.

